STAFF REVIEW · H446 301 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Work growing logarithmically, as with repeated halving.

2. Extra memory beyond input storage.

3. No. It classifies asymptotic growth.

4. Six: 64→32→16→8→4→2→1.

I Do — source worked steps

1. Predict first — Each element is visited once: time O(n).

2. Explain the method — The only additional state is a scalar sum (apart from iterator control): auxiliary space O(1).

We Do — guided CHECK

1. Predict first
Copy performs a constant amount of work n times, so time grows O(n).
Explain the method
New array b stores n values, giving O(n) auxiliary memory.

You Do — source mark schemes

1. O(1) [1], O(n) [1], O(n^2) [1], O(log n) [1].

2. Loop copies n elements [2], O(n) time [1]; new array size n [2], O(n) auxiliary space [1].

3. Bubble O(n^2) worst [2] and O(1) auxiliary in-place [1]; merge O(n log n) worst [2] but typically O(n) auxiliary storage [1].

Exit Ticket — source answer

1. False. Nested loops may use only O(1) additional storage.